

Rahul Kothagundla

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PROFESSIONAL SUMMARY

Software Development Engineer with 1.5+ years building distributed systems at scale. Delivered 70% performance improvements through system re-architecture, maintained 99.9% data integrity across 50K+ daily records and earned 5-star performance rating at TCS. Expertise in backend development, real-time data processing (Kafka, Spark) and cloud infrastructure (Azure, AWS). Seeking SDE role at Amazon to apply distributed systems expertise and customer-obsessed mindset to build products that delight millions.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL, C

Frameworks & Libraries: Apache Kafka, Apache Spark, FastAPI, PySpark, Docker, REST APIs

Databases & Storage: PostgreSQL, Redis, SQL Server, Data Modeling

Cloud & DevOps: Azure Cloud, AWS, Azure Databricks, Docker Compose, CI/CD Pipelines, Git

Core Competencies: Data Structures & Algorithms, Object-Oriented Design, System Design, Distributed Systems

PROFESSIONAL EXPERIENCE

Systems Engineer

June 2024 – Present

Tata Consultancy Services

Hyderabad, India

- Engineered a distributed data processing system using Azure Databricks and Python, re-architecting the computational layer to offload heavy DAX operations from Power BI, reducing system reload time by **70%** and eliminating service disruptions for enterprise stakeholders.
- Designed and deployed production-grade self-monitoring pipeline framework with automated error detection, proactive failure handling and secure backup mechanisms, maintaining **data integrity** across 50K+ daily records and preventing data corruption incidents.
- Built end-to-end ETL pipelines from scratch using Python and SQL, transforming raw Salesforce data into analytics-ready datasets through multi-stage processing across Azure containers, enabling executive-level business intelligence reporting.
- Authored comprehensive technical documentation and developer onboarding framework, reducing new engineer ramp-up time by **90%** (7 weeks → 4 days). Led knowledge transfer to Deloitte team covering architecture, pipelines and production deployment processes. Recognized with **5-star performance rating** in first year.

Machine Learning Engineer

June 2023 – September 2023

Atomstate Technologies Private Limited

Hyderabad, India

- Contributed to high-stakes election analysis project by building predictive models that transformed raw electoral data into strategic insights, providing data-driven recommendations that shaped campaign resource allocation and strategy.
- Developed and deployed a sentiment analysis engine using Natural Language Processing techniques, enabling automated analysis of unstructured text data for business intelligence applications.

Junior Data Scientist

November 2022 – March 2023

Zummit Infolabs

Remote

- Applied advanced NLP techniques to extract actionable insights from ambiguous, unstructured text data, creating analytical models that transformed raw content into structured business intelligence for decision-making.

TECHNICAL PROJECTS

Real-Time Stock Market Streaming System | Python, Apache Kafka, Spark, PostgreSQL, Redis, Docker

2025

- Built production-grade distributed streaming system processing live stock market data with **sub-30-second end-to-end latency**, demonstrating scalable architecture and fault-tolerant design patterns.
- Implemented Kafka-based message queue with 3 partitions for parallel processing, Spark Structured Streaming with 30-second micro-batches and exactly-once semantics and optimized PostgreSQL storage with composite indexes for millisecond query performance.
- Designed multi-tier caching strategy using Redis (5-min TTL) to reduce database load and containerized entire system using Docker Compose for one-command deployment across 9 microservices.
- Deployed live dashboard with real-time data visualization, achieving **99.9% uptime** with automatic health monitoring.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology

Hyderabad, India

Bachelor of Technology in Computer Science Engineering – Data Science, CGPA: 8.55/10

2020 – 2024

CERTIFICATIONS & ACHIEVEMENTS

AWS Cloud Quest: Generative AI Practitioner – Amazon Web Services (January 2026)

Amazon ML Summer School 2023 – Selected for competitive machine learning program by Amazon

Additional Certifications: HackerRank - Problem Solving, ISRO Geo Data Processing with Python